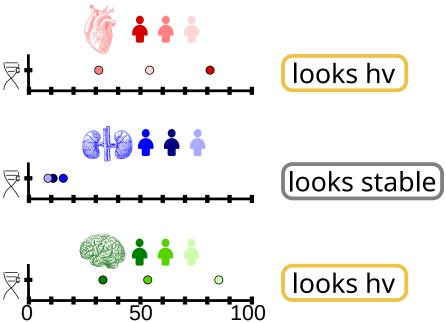
Hierarchical Bayesian model for hvCpG detection

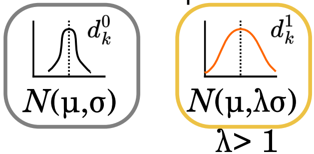
Level 1: the CpG

Is this CpG hypervariable? Z = 1 (yes) with prior probability $\boldsymbol{\alpha_0}$, or Z = 0 (no)

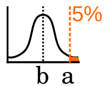
Level 2: the dataset (tissue/cell type)



Each dataset compares two fits



With λ defined as:



$$\lambda = a / b$$

COMBINE (marginalise Z once)

$$P(M|hv) = \alpha_0 \prod_k \left(p_1 d_k^1 + (1 - p_1) d_k^0\right)$$

$$P(M|\overline{hv}) = (1 - \alpha_0) \prod_k \left((1 - p_0) d_k^1 + p_0 d_k^0\right)$$

p_1 true positive

p_0 true negative

Output

Hypervariability score

log ratio of P(M | hv) / P(M | non hv), averaged per-dataset